

Roy Yang

Data-driven senior manager with over 15 years in banking and insurance, including 9+ years in analytics and machine learning. Proven track record delivering deep-dive customer/portfolio analyses, predictive machine-learning models, and executive-level insights that drive acquisition, retention and profit growth.

Expert in Python, Azure/GCP cloud, Power BI & Tableau, with strong stakeholder influence across front-office, risk and product teams.

EXPERIENCE

Data Scientist, Group Treasury

— *AIA Company Limited (secondment by Protiviti), HK*

Sep 2022 - PRESENT

- **Generated US\$2M+ in net new annual premiums** by building a premium-persistence ML model (logistic regression + XGBoost) that surfaced the top 5 drivers of policy lapse and lifted retention 10–15% YoY.

- **Reduced liquidity reporting cycle by 80%** (monthly → daily) by redesigning and automating Liquidity Assessment reports (LQA); insights delivered directly to C-suite via Azure-backed pipelines and an interactive PowerBI dashboard integrating Bloomberg and internal systems.

- **Reduce cash-flow forecast error by >50%** in RSME versus baseline by deploying ARIMA and LSTM models for inflows/outflows, improving budget planning and investment-return optimisation.

- Led a cross-functional analytics project team of 3 analysts and 4 engineers, collaborating with multiple business units to implement analytical strategies and build treasury risk management tools.

Business Strategy Manager, Global Corporate Banking

— *Bank of China (Hong Kong) Limited, HK*

Jan 2017 - Sep 2022

- **Reduced risk-weighted assets (RWA) by >20% while delivering double-digit profit growth** through large-scale deep-dive analyses combining ML and big-data techniques to build quantitative models and data-driven business strategies.

- **Influenced strategic entry into 3 new emerging industries** by producing industry-level research, market-trend analyses, and competitor benchmarks for senior leadership.

- **Achieved >90% adoption among senior managers** for a PowerBI executive dashboard tracking income structure, product

CONTACTS

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PROJECTS

➤ Machine learning forecasting models:

- Developed models for predicting deposit and loan volumes segmented by product type, department, and client profiles.
- Forecasted income by type (interest and fee income) across various products, departments, and divisions.
- Optimized budget planning and enhanced profitability through data-driven financial forecasting.

➤ Client data projects:

- Utilized machine learning for customer segmentation, behavior analysis, and feature analysis.
- Built models for sales and marketing campaign optimization, product recommendations, and client profiling.

➤ Liquidity assessment report:

- Remodeled liquidity calculations for improved accuracy in liquidity measurements.
- Integrated Bloomberg data and connected multiple internal systems through Azure cloud and databases.
- Designed an interactive PowerBI dashboard for real-time portfolio liquidity insights and strategic decision-making.

➤ Business analytic dashboard:

- Developed a comprehensive, interactive dashboard for senior executives to monitor key business metrics (income structure, product volume, financial ratios, etc) across multiple dimensions.
- Provided insights into departmental performance, customer segmentation, and financial metrics to support strategic planning.

volume, and key financial ratios across business lines.

- Built ML models for customer segmentation, campaign optimisation, product recommendation, and client profiling; forecasted deposit/loan volumes and income by product, department, and division.

Senior Officer, Middle Office Personal Banking — *Bank of China (UK) Limited, London*

Feb 2013 - Oct 2015

- Managed project, integrated and implemented the core banking system, internet and telephone banking system, coordinating efforts across IT teams in Europe and China.
- Executed project management tasks including User Acceptance Testing (UAT), risk assessment, and budget forecasting.
- Built a cross-selling database and conducted quantitative data analysis to identify business opportunities in personal and corporate banking.
- Monitored and improved the system capability & staff competence. Developed and revised policy and procedure, Terms & Conditions to control risk and meet compliance standard

Associate Relationship Manager, Premier Banking — *HSBC, HK*

Jun 2010 - Sep 2012

- Achieved profitable growth through value-added financial and investment product sales to high-net-worth (HNW) clients.
- Provided wealth management solutions tailored to individual financial needs, consistently surpassing sales targets.

Personal Financial Manager, Premium Banking — *Standard Chartered Bank (Hong Kong), HK*

Aug 2009 - Jun 2010

- Managed corporate and personal banking portfolios, advising clients on trading strategies and market updates.
- Successfully acquired new clients and increased account profitability through targeted financial solutions.

Personal Banker, Personal Banking Department — *Citibank (Hong Kong) Limited, HK*

Jan 2007 - Apr 2009

- Built and maintained client relationships by providing tailored financial planning solutions.
- Acquired new accounts, driving revenue growth through targeted product cross-selling.

➤ **Premium persistence model:**

- Developed a premium persistence model by applying machine learning to analyze factors like payment methods, policyholder demographics, and product features.
- Enhanced policy retention and profitability through accurate predictive modeling.

TECH SKILL

Programming languages: Python, SQL, R

LLM, machine learning, deep learning: Langchain, PyTorch, Tensorflow

Data analysis, visualization tools: Power BI, Tableau, Excel

Cloud Computing: IBM Watson, Azure, Google Cloud Platform, AWS, Databricks

Statistical packages: SPSS, SAS

Financial terminal: Bloomberg, Wind Financial Terminal

EDUCATION

Master of Science in Computer Science, Financial Computing

The University of Hong Kong (2017-2021)

Master of Science in Risk Management Science

The Chinese University of Hong Kong (2009-2011)

Bachelor of Science in Biology and Psychology

The University of Western Ontario (2002-2006)

LANGUAGES

Fluent in English, Mandarin, and Cantonese